

OTIS ANGB, MA, USA

WMO: 725061

Lat: 41.650N

Lon: 70.517W

Elev: 40

StdP: 100.85

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 14704

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.5	-10.0	-22.2	0.5	-10.8	-19.1	0.7	-7.7	14.6	4.5	13.6	1.7	4.9	300	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	28.9	23.1	27.6	22.0	26.2	20.9	24.2	27.4	23.5	26.0	22.7	24.8	5.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	17.8	25.2	22.7	17.5	24.9	22.3	17.1	24.4	73.5	26.9	69.9	26.1	67.7	24.9	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.8	9.3	DB	-16.6	31.5	4.2	1.7	-19.6	32.7	-22.0	33.8	-24.4	34.7	-27.4	36.0	(4)
(5)			WB	-17.5	25.4	3.9	1.1	-20.2	26.2	-22.5	26.8	-24.6	27.4	-27.4	28.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-0.5	-0.2	2.6	7.9	13.4	17.5	21.8	21.1	17.7	12.6	7.0	2.8
	DBStd	8.74	5.07	4.74	4.44	3.71	3.51	3.01	2.54	2.58	3.41	4.05	4.33	4.94
	HDD10.0	1294	327	286	232	83	8	0	0	0	0	20	109	229
	HDD18.3	3177	584	518	488	313	157	51	3	5	52	182	341	483
	CDD10.0	1427	1	1	3	20	114	224	365	346	231	101	18	4
	CDD18.3	268	0	0	0	0	5	25	109	92	33	4	0	0
	CDH23.3	1284	0	0	0	3	32	117	547	456	121	8	0	0
	CDH26.7	231	0	0	0	0	5	19	114	79	13	1	0	0
Wind	WSAvg	4.4	4.8	5.0	5.0	4.8	4.4	4.1	3.9	3.6	3.9	4.5	4.6	4.5
Precipitation	PrecAvg	1188	98	83	123	110	97	85	70	95	110	94	107	122
	PrecMax	1511	159	185	261	201	169	280	186	192	326	253	175	240
	PrecMin	986	47	45	15	38	30	9	19	8	33	21	18	38
	PrecStd	128	33	36	55	40	36	62	36	44	68	59	46	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	13.0	17.4	22.4	27.1	29.2	31.3	30.2	28.0	24.1	18.8	17.0
		MCWB	12.9	10.5	12.8	15.5	20.1	22.2	24.6	24.1	22.7	19.0	16.8	15.7
	2%	DB	11.2	11.0	12.9	17.9	23.0	26.3	29.0	28.7	26.2	22.2	17.2	13.9
		MCWB	10.6	9.8	9.7	13.5	17.8	19.5	23.3	22.8	21.1	19.0	15.4	12.9
	5%	DB	9.0	8.8	10.8	15.9	21.2	24.7	27.7	27.4	24.8	20.0	15.2	12.4
		MCWB	8.1	7.4	8.4	11.4	16.3	18.8	22.5	21.9	20.9	18.1	13.3	11.3
	10%	DB	7.4	7.1	8.8	13.7	18.9	22.8	26.3	26.0	22.9	18.8	13.9	10.2
		MCWB	6.0	5.4	6.9	9.8	15.0	18.3	21.3	21.1	19.9	17.3	12.0	9.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	12.4	13.7	17.6	21.1	23.1	25.5	25.3	24.0	21.2	17.6	15.9
		MCDB	13.4	12.9	15.5	19.5	25.7	28.6	29.8	28.7	26.9	22.3	18.0	16.4
	2%	WB	10.5	9.8	10.8	15.1	19.0	21.6	24.4	24.3	22.9	19.7	15.9	13.2
		MCDB	10.8	10.2	12.6	17.1	22.2	24.8	27.9	27.1	24.7	20.6	16.7	13.8
	5%	WB	8.0	7.7	8.8	12.7	17.5	20.6	23.6	23.5	22.2	18.6	14.2	11.4
		MCDB	8.4	8.4	10.0	14.9	19.1	22.8	26.3	25.8	23.9	19.7	15.4	12.1
	10%	WB	6.1	5.2	7.1	11.0	16.2	19.6	22.9	22.7	21.1	17.3	12.5	9.3
		MCDB	7.3	6.6	8.2	12.6	18.5	21.6	24.9	24.6	22.6	18.2	13.7	10.0
Mean Daily Temperature Range	5% DB	MDBR	7.8	8.0	7.9	8.8	8.4	8.0	7.7	8.1	8.4	8.5	8.1	7.7
		MCDBR	10.1	10.1	10.1	11.7	11.4	10.5	9.0	9.2	9.3	9.2	9.3	9.4
		MCWBR	10.1	9.0	7.6	7.3	6.4	5.6	4.3	4.7	5.6	6.5	8.0	9.6
	5% WB	MCDBR	9.6	10.0	9.6	10.3	9.7	8.5	7.9	7.6	7.8	7.9	8.1	9.4
		MCWBR	10.0	9.8	8.1	7.8	6.6	5.4	4.3	4.3	5.3	6.6	8.1	9.8
Clear-Sky Solar Irradiance	taub	0.307	0.313	0.336	0.373	0.409	0.460	0.483	0.437	0.373	0.345	0.317	0.305	
	taud	2.513	2.500	2.474	2.394	2.337	2.230	2.173	2.292	2.451	2.533	2.568	2.563	
	Ebn at Noon	849	900	915	898	869	822	797	825	863	852	832	823	
	Edh at Noon	73	85	98	114	124	139	145	125	100	82	69	64	
All-Sky Solar Radiation	RadAvg	1.78	2.67	3.68	4.75	5.34	5.74	5.94	5.34	4.33	2.95	2.00	1.50	
	RadStd	0.14	0.26	0.28	0.51	0.52	0.53	0.33	0.28	0.30	0.25	0.14	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+45	(47)

Nomenclature: See separate page